MILITARY SPECIFICATION

BOILER, STEAM, HIGH PRESSURE, WATER TUBE TYPE, AUXILIARY-NAVY SHIPBOARD

1. SCOPE

1.1 This specification covers packaged type oil-fired natural-circulation water-tube type boilers of 2-drum, single uptake design with burner, all necessary appurtenances, controls, and auxiliary equipment mounted as an integral unit on a steel base so as to make a complete, self-contained steam generating unit.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods Of
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-C-915 - Cable, Cord and Wire, Electrical (Shipboard Use)
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use
MIL-R-2033 - Relays for Naval Shipboard Electrical Service
MIL-E-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard
MIL-C-2174 - Controllers, Direct-Current, Naval Shipboard
MIL-C-2194 - Cables, Power, Electrical, Reduced Diameter Type, Naval Shipboard.
MIL-C-2212 - Controllers, Alternating-Current, Naval Shipboard
MIL-B-2678 - Brushes, Wire, Boiler-Tube, (Expanding Type for Use With Cleaning-Outfits, Boiler Tube, Electric and Air-Motor-Driven)
MIL-C-5015 - Connectors, Electric, "AN" Type
MIL-J-5624 - Jet Fuel, JP-4 and JP-5
MIL-P-14276 - Paint, Heat-Resisting; Silicone, Aluminum
MIL-P-15024 - Plates, Identification-Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment
MIL-M-15071 - Manuals, Equipment and Systems
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-P-15384 - Plastic Mix, Refractory (Water-Wall, Boiler, Chrome Ore Plastic).
MIL-B-16008 - Brick, Insulating, High Temperature, Fire Clay
MIL-I-16165 - Interference Shielding, Engine, Electrical Systems
MIL-T-16286 - Tubes, Steel, Seamless, Marine Boiler Application
MIL-P-16298 - Preservation, Packaging, Packing, and Marking of Electric Machines Having Rotating Parts and Associated Repair Parts
MIL-W-16878 - Wire, Electrical, Insulated, High Temperature
MIL-F-16884 - Fuel Oil, Diesel, Marine
MIL-I-16910 - Interference Measurement, Radio, Methods and Limits; 14 Kilocycles to 1000 Megacycles
MIL-M-17059 - Motors, Alternating-Current, Fractional HP (Shipboard Use)
MIL-M-17060 - Motors, Alternating-Current, Integral Horsepower (Shipboard Use)
MIL-T-17188 - Tubes, Carbon Steel, Resistance Welded, Marine Boiler
MIL-P-17286 - Propulsion and Auxiliary Steam Turbines and Gears (Includes Repair Parts, Tools, Accessories and Instruments); Packaging of
MIL-T-17370 - Tube and Flue Cleaner, Boiler, Internal, Air-Motor-Driven

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MILITARY (Cont'd)
MIL-M-17413 - Motors, Direct-Current, Integral HP, Naval Shipboard
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-M-17556 - Motor, Direct-Current, Fractional HP (Shipboard Use)
MIL-P-17945 - Pumps, Centrifugal, Peripheral Type, Miscellaneous Naval Shipboard Service
MIL-F-19004 - Fans, Centrifugal, Fixed and Portable, Ventilation, Naval Shipboard

STANDARDS

MILITARY
MIL-STD-22 - Welded Joint Designs
MIL-STD-130 - Identification Marking for U.S. Military Property
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment
MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy

DRAWINGS

BUREAU OF SHIPS
810-1385850 - Piping Gages for all Services

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N.Y.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS
Boiler Construction Code

(Application for copies should be addressed to the American Society for Mechanical Engineers, 29 West 39th St., New York, N.Y.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS. -

3.1 Preproduction unit. - Prior to the beginning of production one boiler, complete with auxiliaries, shall be submitted for the preproduction inspection specified in 4.2 (see 6.3).

3.2 Materials. - Except as otherwise specified herein, all materials shall conform to those specified in the ASME Boiler Construction Code except as specified in 3.2.1 through 3.2.10.

3.2.1 Cast iron shall not be used in any part of the boiler or any part of the auxiliaries and accessories or controls. Nodular (ductile) iron castings are permissible subject to written approval by the bureau or agency concerned.

3.2.2 All valves, including boiler safety valves, shall conform to the requirements of the ASME Boiler Construction Code except that the safety valves shall be cast or forged steel and all other valves shall be bronze or cast or forged steel.

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3.2.3 Material for boiler tubes shall be seamless steel in accordance with MIL-T-16286 or electric resistance welded steel in accordance with MIL-T-17188 or the commercial equivalents (see 3.2.9).

3.2.4 Furnace brickwork shall conform to MIL-B-16008 or the commercial equivalent (see 3.2.9) or shall be a product which is listed on or which has been approved for listing on Qualified Products List QPL-15606. Only Navy standard shapes will be acceptable.

3.2.5 Castable hydraulic setting refractory used to form the burner cone opening shall be a product which is listed on or which has been approved for listing on Qualified Products List QPL-717.

3.2.6 Air setting mortar used for laying brickwork shall be a product which is listed on or which has been approved for listing on Qualified Products List QPL-15842.

3.2.7 Plastic water wall refractory, when used, shall be in accordance with MIL-P-15384 or the commercial equivalent (see 3.2.9).

3.2.8 Thermal block insulation shall be a class 3 or 4 product listed on or which has been approved for listing on Qualified Products List QPL-2819.

3.2.9 When commercial equivalents are supplied, they shall be identified by commercial trade name and physical and chemical properties shall be given.

3.2.10 The boiler shall be equipped with necessary brickwork, insulation, and lagging consisting of a metal jacket over the steam and water drum insulation.

3.3 Design. -

3.3.1 Except as otherwise specified herein, the boiler shall be designed in accordance with the ASME Boiler Construction Code.

3.3.2 The boiler shall be oil-fired natural-circulation water-tube type of 2-drum single uptake design. The hand of the boiler shall be as defined and specified (see 6.1).

3.3.3 The boiler shall be designed to permit normal operation with a momentary roll up to 45 degrees, or a permanent list of 15 degrees to either side with a permanent inclination of 5 degrees fore and aft with boilers installed with longitudinal axis of the steam drum in fore and aft direction.

3.3.4 The boiler shall have the following design and operating characteristics:

(a) Pressure design	150 pounds per square inch (p.s.i.)
(b) Pressure, working	125 p.s.i.
(c) Pressure, safety valve popping	145 through 150 p.s.i.
(d) Capacity, saturated steam, pounds per hour, actual evaporation at 125 p.s.i. working pressure with feed water temperature at 100° F.:	
(1) Normal rated capacity - As specified (see 6.1).	
(2) Maximum rated capacity - As specified (see 6.1).	
(e) Exhaust gas temperature	650°F. maximum at rated capacity.
(f) Efficiency	75 percent minimum at maximum capacity.
(g) Fuel burned	JP-5 jet fuel in accordance with MIL-J-5624 or Diesel fuel oil, in accordance with MIL-F-16884 as specified (see 3.6.2 and 6.1).

3.3.5 The boiler shall be designed for automatic feed water regulation (see 3.7.4.1) and for either manual or automatic burner operation as specified (see 6.1). If automatic burner operation is specified, provision

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shall be made for by-passing the automatic burner controls in order to operate the burner manually whenever necessary. Provisions shall be made in the feed water system for by-passing the automatic regulator and controlling the boiler feed water manually when automatic feeding is not used.

3.3.5.1 The manual burner operation shall consist of electric spark ignition, flame failure control, and manual regulation of fuel oil and air supplies. Ignition shall be by electric arc (electrodes) manually controlled by push button. The circuit shall be wired such that fuel admission follows 5 seconds after the push button is depressed and the ignition circuit is energized. If combustion does not take place, the oil solenoid valve shall close within 5 seconds plus or minus one second. Electrodes shall be locked securely in place to prevent movement.

3.3.5.2 The automatic burner operation shall feature automatic spark ignition, flame failure control, and automatic modulation of firing rates within a range corresponding to boiler steaming loads from the maximum rated capacity to 10 percent of the maximum rated capacity, and automatic "On and Off" cycling at steaming loads below 10 percent of the maximum rated capacity (see 3.3.4). Ignition shall be by electric arc (electrodes) automatically controlled by programmed sequence (see 3.7.2.1.1). Electrodes shall be locked securely in place to prevent movement.

3.3.5.2.1 Working pressure for boilers equipped for automatic burner operation shall be approximately the mean between the automatic burner cut-in and cut-off pressures.

3.3.6 The boiler shall be equipped with all auxiliaries, accessories, and controls to make a self-contained and complete steam generating unit when connected with electric power supply and fuel oil and feed water supply and return lines to be provided by the installing activity. No economizers, air heaters or superheaters shall be fitted to the boiler.

3.3.6.1 The boiler and all auxiliaries shall be adequately supported on a base. Means shall be provided for connecting the base to the ship's deck or foundation.

3.3.6.1.1 Shock resistance. - The base, all support members for boiler and for auxiliaries and hold down bolts shall be designed to withstand the shock loadings shown on figure 1. Shock stresses shall be considered for each direction separately and shall not be combined. Bolts stressed in shear shall be installed in holes of the following sizes:

Nominal bolt diameter	Diameter of hole
Up to and including 3/4 inch	Nominal bolt diameter plus 1/32 inch
Larger than 3/4 inch	Nominal bolt diameter plus 1/16 inch

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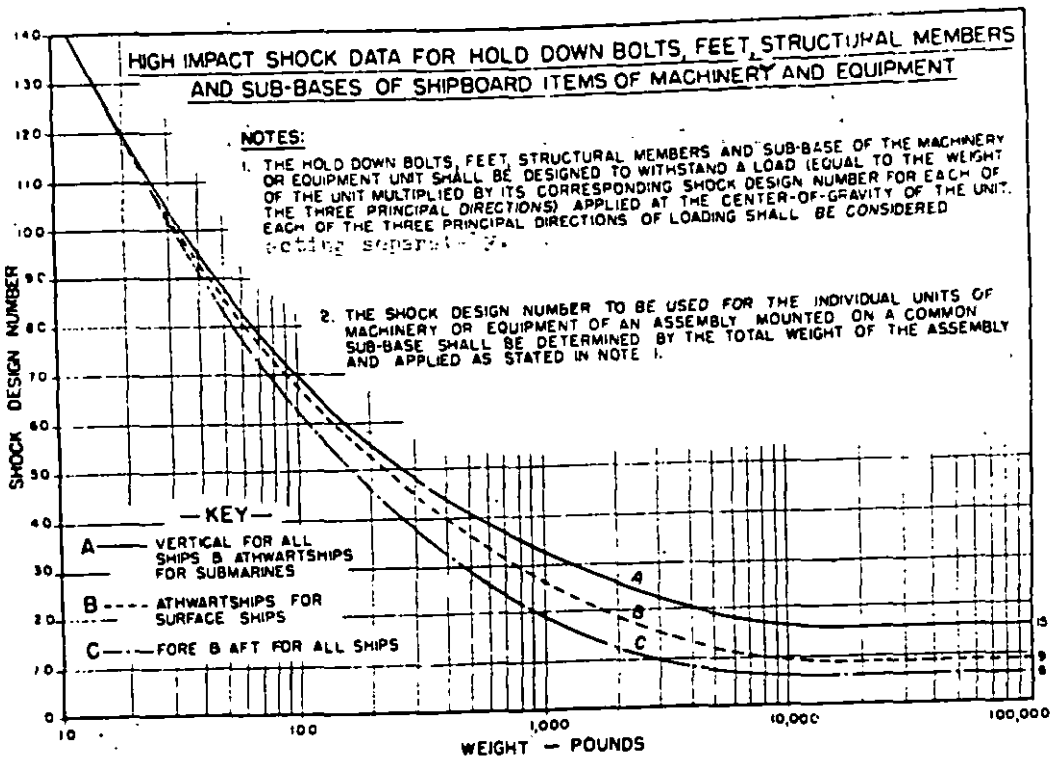


Figure 1 - Shock

3.3.6.1.2 Auxiliaries and other equipments rigidly supported shall not be attached to 2-structures which can deflect relative to each other, under shock loading.

3.3.6.2 The auxiliaries, accessories and controls shall include the following:

(a) Burner (see 3.3.5 and 3.7.1).
(b) Ignition system (see 3.3.5.1 and 3.7.2.1).
(c) Flame failure safety control (combustion safeguard) (see 3.7.2).
(d) Fuel oil system (see 3.7.3).
(e) Blower and air regulating mechanism (see 3.7.2.3 and 3.7.8).
(f) Feed water system (see 3.7.4).
(g) Automatic feed water regulator (see 3.7.4.1 and 3.3.5).
(h) Automatic low water and high temperature limit cut-off control (see 3.7.2.4).
(i) Motors and controllers as required by the driven auxiliaries (see 3.7.7).
(j) Soot blowers as required (see 3.7.5).
(k) Smoke indicator (see 3.7.6).
(l) Safety valves, as required (see 3.2.2).
(m) Steam stop valve.
(n) Bottom blow off valves.
(o) Surface blow off valve.
(p) Salinometer valve.
(q) Air vent valve.
(r) 2-Water gages, with valves (10 inches visibility each, reflex flat glass for size No. 7 glasses), each gage separately mounted.
(s) Steam pressure gage, 0-300 No. 8-1/2 inches dial.
(t) Combustion chamber observation window and manual ignition port of swing type (see 3.7.1.2).

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3.3.6.3 Automatic controls, except for motor starters, shall be suitable for operation with 220 volt a.c. current. A suitable transformer shall be included on the boiler control panel if the available power supply is other than 220 volts a.c. (see 6.1). A rotary inverter shall be provided when power source is d.c.

3.3.6.4 Drives for auxiliary equipment. - Wherever practicable, the drive and driven shall be direct coupled with flexible coupling. Belt drives may be employed subject to obtaining specific approval from the bureau or agency concerned. Belt drives when employed shall conform to the following:

(a) Matched belts, as a set, shall be furnished for multiple belt drives.
(b) Belt speeds in excess of 5,500 feet per minute are not acceptable.
(c) Drive and driven sheaves shall be in balance (statically and dynamically), as installed, in accordance with internally excited vibration, type II requirements, of MIL-STD-167.
(d) Axial movement in drive and driven shall be controlled by thrust bearings. Thrust bearings may be of the combined radial and thrust rolling contact type.

3.3.6.5 The boiler shall be equipped with necessary brickwork, insulation and lagging consisting of a metal jacket over the steam and water drum insulation.

3.4 Construction. - Except as otherwise specified herein, the boiler shall be constructed in accordance with the ASME Boiler Construction Code.

3.4.1 The boiler steam drum shall be of fusion welded construction. Welding and inspection shall be in accordance with MIL-STD-278, class A1 pressure vessels and MIL-STD-22.

3.4.1.1 The inside diameter of the steam drum shall not be less than 24 inches. The manhole in the steam drum and the water gages shall be located on the front end of the boiler. The manhole shall be elliptical 12 inches by 16 inches. The steam drum shall be equipped with the following internal fittings:

(a) Dry pipe.
(b) Surface blow pipe.
(c) Feed pipe.
(d) Steam baffles.
(e) Swash plates.

3.4.1.2 Unstayed dished and flanged heads with pressure on concave side shall be used for steam and water drums and headers, except that integral flat heads or flanged plates conforming to figure P-21 (b) and figure P-21 (c), respectively, of the ASME Boiler Construction Code shall be acceptable for the water drums and headers of inside diameter smaller than 14 inches. The heads shall be connected to drums or headers wherever practicable by means of a butt joint welded on both sides.

3.4.2 Boiler tubes shall conform to the dimensions shown in table I for outside diameters and wall thicknesses:

Table I - Dimensions

Outside diameter (inches)	Wall thickness (inches)
3/4 and 1	0.085
1-1/8 and 1-1/4	.109
1-3/8 and 1-1/2	.120
1-3/4 and 2	.134
2-1/4, 2-1/2, and 2-3/4	.165

3.4.2.1 No studded tubes shall be permitted in the construction of the boiler. Finned tubes may be used.

3.4.3 Provisions shall be made for connecting the air supply duct from the weather to the forced draft blower inlet or to the air admission opening in the air casing and for connecting boiler uptake to the flue gas outlet. The blower shall be located as high as possible on the boiler (on uptake side or furnace side as convenient) and air for combustion shall pass downward through double casing into burner plenum.

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3.4.4 The casing, including air casing if used, shall be of steel plate and welded construction except the uptake side casing and all removable doors and panels shall be bolted. The casing (both inner and outer for air-encased boilers) shall be gas and airtight and shall be designed to withstand air and gas pressures under maximum operating conditions. Necessary provisions shall be made for expansion. The following features shall be provided in the casing:

(a) Furnace access door in rear casing.
(b) At least 2 air lance openings at suitable locations in the front casing.
(c) A peep-hole in the rear casing.
(d) Complete pressurized double casings or front casing windbox.
(e) Flanged outlet at boiler breeching.

3.4.4.1 All fittings, nozzles, and pipes which pass through the casing shall be tight against leakage; stuffing boxes or other seals shall be provided as required and necessary provisions shall be made for expansion.

3.4.4.2 Two coats of silicone paint conforming to MIL-P-14276 shall be given to both sides of the air casing, to the outside surfaces of inner casing, to the steam and water drums, and to the drum lagging.

3.4.4.3 All pressure gages shall be flush panel-mounted, gage piping conforming to Drawing 810-1385850.

3.4.4.4 Shock hardening of brickwork. - All areas of plastic or castable refractory material shall be anchored, no matter how small. In refractory areas around peepholes and sootblowers, at least one anchor strip shall be installed. Areas of formed refractory in front wall burners, and in rear walls, under steam drum, over water drum, and in corners, shall be anchored. Floor brickwork shall be anchored and brick-pans strengthened accordingly. Castable or plastic baffles shall be reinforced with nichrome wire intertwined around tubes. All corbels and fillets shall be anchored to headers or casings being protected. Nichrome wire shall be intertwined around anchors, tubes, and through corbels for reinforcement.

3.5 Maintainability

3.5.1 The component parts of the boiler assembly shall be arranged with respect to each other to permit easy access for servicing or repair and replacement of parts.

3.5.2 The water drum shall have a sufficient number of handholes to permit easy inspection, expanding and cleaning of boiler tubes.

3.5.3 The configuration of all boiler tubes shall be such as to permit mechanical cleaning of watersides using air driven tube cleaning equipment conforming to MIL-C-17370.

3.6 Performance. - The boiler performance shall be such as to conform to the design requirements specified in 3.3.3, 3.3.4, and 3.6.1 through 3.6.3.

3.6.1 Moisture content in steam shall not exceed 1 percent when boiler is steaming at the normal rated capacity with the minimum chloride and total solids concentration in boiler water of 550 and 1175 parts per million, respectively.

3.6.2 Performance of the boiler shall be based on burning Diesel fuel oil of the following analysis conforming to MIL-F-16884 or on burning JP-5 fuel conforming to MIL-J-5624:

Carbon	85 percent.
Hydrogen	14 percent.
Sulphur	1 percent.
Nitrogen	Nil.
Oxygen and undermined	Nil.
Moisture	Trace.
B.t.u./pound	19,375 for diesel fuel; 19,500 for JP-5 fuel.

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3.6.3 All parts of the boiler assembly shall be insulated so that at overload operation, the average surface temperature of casing shall not exceed 135°F. (front, rear and each side considered separately), considering ambient air taken one foot away from the casing at a temperature of 100°F. and a velocity of 0.5 feet per second. Requests for waivers of temperature spots in excess of 145°F. due to connections or structural ties shall be submitted to the bureau or activity concerned for approval.

3.7 Details of components. -

3.7.1 Burner. - The burner shall be a wide range mechanical pressure, steam pressure, or air pressure atomizing type, as approved by the bureau or agency concerned. The burner shall be capable of smokeless operation at all firing rates corresponding to boiler steaming loads from the maximum rated capacity to 10 percent of the maximum rated capacity. A turndown range of 20 percent of the maximum rated capacity shall be without change of the sprayer plate.

3.7.1.1 When air pressure atomizing burner is furnished, an automatic safety cut-off control designed to stop burner operation in case of air supply failure shall be provided with the burner.

3.7.1.2 An observation window and a manual ignition port shall be installed on the burner. For both manual and automatic operation, provision shall be made for emergency torch ignition in addition to the electric spark ignition.

3.7.1.3 An air compressor when required for an air pressure atomizing burner shall be furnished and installed within the space limitations specified for the complete boiler assembly.

3.7.2 Controls. - The controls shall conform to the design requirements specified in 3.3.6.3 and 3.3.6.4 and shall consist of the following as required for manual or automatic control operation:

(a) Automatic ignition and combustion control (automatic operation only).
(b) Manual ignition (see 3.3.5.1).
(c) Flame failure safety control.
(d) Air regulating control.
(e) Automatic low water and high temperature limit cut-off control.
(f) Feed water regulator.
(g) Pressure limit control.

3.7.2.1 Automatic modulation. - The automatic ignition and combustion control system shall be designed to automatically modulate the burner without "On" and "Off" cycling or change of sprayer plates through its entire firing range when the boiler is steaming at variable loads between the maximum rated capacity and 20 percent of the maximum rated capacity, and to automatically cycle the burner "On" and "Off" when the boiler is steaming at loads less than 20 percent of the maximum capacity.

3.7.2.1.1 The automatic ignition and combustion control system shall be arranged for the following sequence of operation upon the initial start and all subsequent automatic restarts during "On" and "Off" operation:

(a) A pre-purge period of 45 seconds with wide open damper.
(b) Automatic ignition.
(c) Low fire or start-up position interlock in automatic ignition circuit.
(d) Air pressure safety switch interlock in burner circuit.
(e) Steam pressure limit interlock.
(f) Interlock preventing delivery of fuel to burner until start-up or pilot flame is proved by combustion safeguard (see 3.7.2.2).
(g) Interrupted ignition circuit.
(h) Post-purge period of 30 seconds with wide open damper.
(i) Safety shut down on ignition failure and on flame failure (see 3.7.2.2).
(j) The overall duration of the programming cycle shall not exceed 120 seconds.

3.7.2.1.2 The automatic ignition and combustion control system shall be arranged to shut off the fuel oil solenoid valve within 5 seconds plus or minus 1 second after opening in case of ignition failure. Following this shutoff of the fuel oil solenoid valve the blower motor shall continue operation for at least 30 seconds to purge the combustion chamber. During this period the burner shall move to the high fire position. After

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completion of the purging period a safety shut down of the burner operation shall take place. Upon such safety shut-down the burner shall be restarted only by manually operating the reset button on the control panel. Means shall be provided to drain possible solenoid valve fuel oil leakage outside boiler casing.

3.7.2.1.3 Timing periods of the automatic ignition and combustion shall not vary due to ambient temperature variation between 32°F. and 120°F. or due to supply voltage variation of plus or minus 15 percent of the nominal voltage. In case of failure to supply voltage the programming control shall automatically recycle following the specified sequence when power is restored.

3.7.2.2 The flame failure safety control may be of photo-electric or photo-conductive type. The control shall readily detect flame under all firing conditions and instantly close the fuel oil solenoid valve in case of flame failure of 2 to 4 seconds duration. Following this shutoff of the fuel oil solenoid valve, the blower motor shall continue operation for at least 30 seconds to purge the combustion chamber. During this period the burner shall move to the high fire position. After completion of the purging period, a safety shut-down of the burner operation shall take place. Upon such safety shut-down, the burner shall be restarted only by manually operating the reset button on the control panel. When manual operation is specified (see 3.3.5) following the shutoff of the oil solenoid valve, the blower shall continue operation until manually secured.

3.7.2.3 An air regulating control (damper or louvers) shall be provided for controlling air supply to the burner in correct proportion to its firing rates. The control shall be interlocked with the metering oil valve in the fuel oil system. The linkage shall be designed to provide simultaneous regulation of the fuel oil and air supplies to the burner when boiler is steaming at variable loads. This control shall be hand operated on manual boilers and motor operated on automatic boilers. Means shall be provided in the linkage for manual adjustment of fuel oil-air ratio as necessary for smokeless operation of the burner throughout its entire firing range.

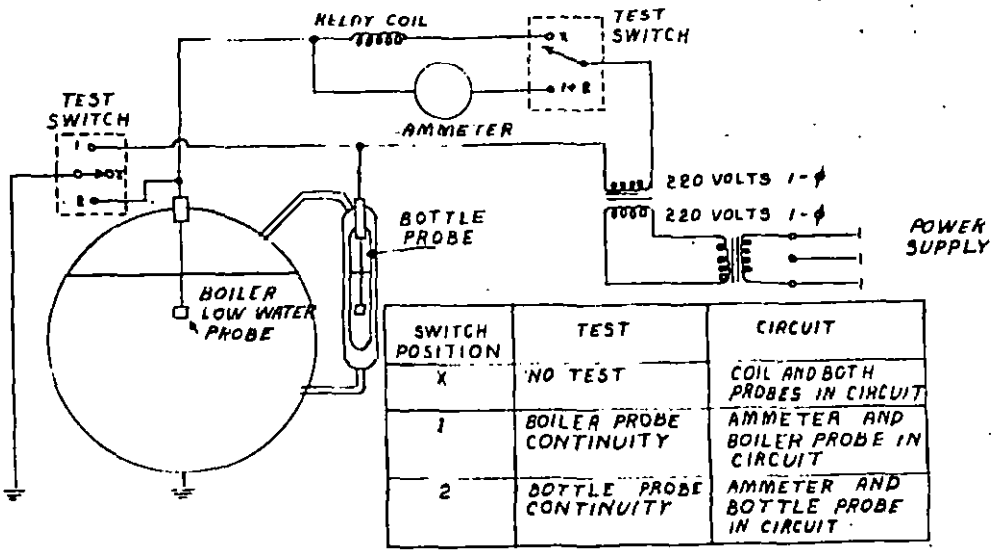
3.7.2.4 The automatic low water and high temperature limit cut-off control may be either a float type or an electrode probe type, and shall be designed for mounting on the top of the steam drum at a location equidistant from the drum ends. The control shall be arranged to function without short cycling due to shifting of water in the steam drum. The control shall cut-off the burner operation in the event the lowest permissible water level is reached. After this emergency cut-off the boiler shall be restarted only by manually operating the push button master switch on the control panel after water level has been regained.

3.7.2.4.1 Float type. - The float type shall be designed for flange mounting and shall have a float cage or surge chamber surrounding the float. The unit shall be removable through a companion opening in the drum. The switch mechanism shall be mechanically or magnetically operated. The float control shall be a unit which has passed Hi-shock and environmental vibration requirements in accordance with MIL-S-901 and MIL-STD-167, respectively.

3.7.2.4.2 Electrode probe type. - The electrode probe type designed for flanged mounting shall be comprised of an electrode plug and probe extension. The probe extension shall be connected to the electrode plug and locked securely in place to prevent backing off. The probes shall be located and arranged to form an isolated ungrounded circuit. Each probe shall be mounted independently. Probes complete with cage or surge chamber shall be removable through companion opening in drum. The electrodes shall be insulated with teflon. A ground test circuit shall be wired into the control circuit to permit periodic check of electrodes for possible grounded condition. Arrangement shall be similar to that shown in figure 2. Switches and indicators shall be located on the control console operating panel (see 3.7.2.5) and shall be units that have passed Hi-shock and environmental vibration tests in accordance with MIL-S-901 and MIL-STD-167, respectively. Calculations shall be made to determine the ammeter test reading (in m.a.) that will indicate probe grounding of sufficient intensity to pass ground current equal to the relay coil drop-out current. This ammeter reading when adjusted by a safety factor of four should be the indicated ground point as shown on the face of the ammeter scale. Calculations shall be submitted to the bureau or agency concerned for review and approval. A time delay relay shall be provided in the low water cut-off circuit to prevent nuisance shut-downs of the boiler due to uncovering of the probes caused by ship's roll.

3.7.2.5 All controls shall be mounted in a class 1 dripproof enclosure, in accordance with MIL-E-2036. The enclosure shall be mounted on an operating panel consolidating all indicators, lights and push buttons thereon. The service of each push button shall be clearly marked.

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SWITCH POSITION	TEST	CIRCUIT
X	NO TEST	COIL AND BOTH PROBES IN CIRCUIT
1	BOILER PROBE CONTINUITY	AMMETER AND BOILER PROBE IN CIRCUIT
2	BOTTLE PROBE CONTINUITY	AMMETER AND BOTTLE PROBE IN CIRCUIT

Figure 2 - Ground indication circuit

3.7.3 Fuel oil system. -

3.7.3.1 The fuel oil system shall include a fuel oil strainer, fuel oil pump, fuel oil filter, relief valve, solenoid oil valve, metering oil valve, supply and return line pressure gages, interconnecting valves and fittings, and pilot burner (if used).

3.7.3.2 The fuel oil pump shall be of the self priming type and shall have a minimum suction lift of 12 feet. The pump shall be capable of supplying all requisite oil and shall have characteristics suitable for operation with the burner furnished. The pump shall have a direct coupled transmission and a separate motor drive. The motor shall have no other loading.

3.7.3.3 Fuel oil strainers and filters shall be either the duplex type or the simplex self cleaning type designed to prevent interruption in flow when cleaning.

3.7.3.4 When automatic operation is specified (see 3.3.5) the fuel oil system shall be arranged to permit manual regulation of the burner throughout the entire firing range when automatic operation is prevented.

3.7.4 Feed water system. -

3.7.4.1 The feed water system shall include a water strainer, feed pump, relief valve, feed water treatment tank, pressure gage, feed water regulator, feed water stop and check valves, and necessary connecting piping, valves and fittings.

3.7.4.2 The feed water regulator shall be designed to automatically flow through the tubes at all steaming loads. The steam drum water level shall hold within plus or minus 4 inches at all steaming rates and during on-off operation.

3.7.4.3 A feed water treatment tank shall be valved in the by-pass line. The tank shall be of one gallon capacity or other capacity as approved by the bureau or agency concerned. The tank shall be equipped with a funnel for adding chemical solutions, a vent valve, and a drain valve.

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3.7.4.4 The boiler feed pump shall be centrifugal or positive displacement type capable of feeding requisite water to the boiler when steaming at maximum capacity at a boiler pressure of 150 pounds per square inch gage (p.s.i.g.). The pump shall be designed to be self priming. Pump discharge head shall have a practical margin when injecting water into the boiler through a feed line containing a feed water regulator control valve, and stop and check valves. Design and construction shall be in accordance with MIL-P-17945. The pump shall be motor driven direct coupled with no other loading on the motor. The pump shall be horizontally mounted unless otherwise approved.

3.7.5 Soot blowers. - The soot blowers shall be of a type that has been approved for Naval service. The soot blowers shall be installed on the front of the boiler.

3.7.6 Smoke indicators. - Smoke indicators will be required for all manually operated boilers and for all boilers of 10,000 pounds per hour normal capacity and above. Smoke indicators shall be of a periscope type and shall be in accordance with MIL-I-2002. The smoke indicator shall be shipped with the boiler for installation in the boiler uptake by the installing activity.

3.7.7 Electrical equipment. -

3.7.7.1 Electric motors. - Electric motors for driven auxiliaries shall be in accordance with MIL-M-17413, MIL-M-17060, MIL-M-17556 or MIL-M-17059 as required.

3.7.7.1.1 Ship's electric power supply available shall be as specified (see 6.1).

3.7.7.1.2 Motors shall conform to the following classification requirements:

Horsepower	As required.
Volts	As required.
Number of phases (a.c. only)	As required.
Frequency (a.c. only)	As required.
Speed	Constant.
Duty	Continuous.
Enclosure	Dripproof protected.
Insulation class	A, B or H.
Service	Navy A.
Ambient temperature	50°C.
Type	Induction (for a.c. motors).
Winding	Squirrel cage (for a.c. motors).
Bearing	Ball.
Sub-classification	Design B (for a.c. motors).
Radio noise suppresion	Required (for d.c. motors).
Winding	Shunt (for d.c. motors).

3.7.7.2 Rotary inverter. - The rotary inverter when used to supply alternating current power for the ignition system shall be in accordance with MIL-M-17556. The alternating current take off shall be 60 cycle, single phase, 220 volts. Inverters shall conform to the following classification requirements:

Prime mover	D.c. motor.
Size (KW)	As required.
Type	Revolving armature.
Enclosure	Dripproof protected.
Speed	Constant.
Duty	Continuous.
Service	Navy A.
Bearing	Ball.
Ambient temperature	50°C.
Insulation	A, B or H.
Radio noise suppression	Required.

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3.7.7.3 Modulating motor. - The modulating motor shall conform to commercial marine standards and to the following requirements:

Timing 15 or 30 seconds	As required.
Stroke travel	As required.
Load torque (foot pounds)	150 percent of calculated load.
Voltage	As required by circuit design.
Location	Remote.
Enclosure	Dripproof.

3.7.7.4 Controllers. - Electric controllers (motor starters) shall be mounted remote to the boiler control circuit enclosure and shall be in accordance with MIL-C-2174 for d.c. controllers and in accordance with MIL-C-2212 for a.c. controllers.

3.7.7.4.1 The controllers shall conform to the following requirements:

Service	Navy A.
Duty	Continuous.
Rating	As required.
Enclosure	Dripproof.
Operation	Magnetic.
Type	Resistor (for d.c. controllers); across line (for a.c. controllers).
Function	Motor starting.
Protection	Overload protection with emergency run feature and under-voltage protection.
Performance	As required.
Master switch	Remote, spray tight, 3-element push button marked "Start", "Stop" and "Emergency Run".

3.7.7.5 Relays. - All relays, except time delay relays, shall be high impact shockproof units in accordance with MIL-R-2033 and the following:

Type	As required.
Tripping characteristics	As required.
Voltage	As required.
Duty	Continuous or intermittent as required.
Ambient temperature	50°C. (122°F.).

3.7.7.6 Switches. - Switches and push buttons shall be Hi-shock design and shall comply with requirements of MIL-C-2212 and the following:

Voltage rating	As required.
Location	Mounted on operating panel.
Type	Push button or rotary.
Ambient temperature	50°C. (122°F.).

3.7.7.6.1 Limit switches shall have a remote location mounted near the parent machine. Their enclosure shall be dripproof.

3.7.7.7 Pressure switches. - All pressure switches except the modulating pressure control switch shall be Hi-shock design and shall comply with requirements of MIL-C-2212 and the following:

Voltage rating	As required.
Location	Remote.
Type	Pressure actuated.
Ambient temperature	As required (50°C. (122°F.) min.
Enclosure	Dripproof.

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3.7.7.8 Solenoid valves. - Solenoid valves shall be Hi-shock tested in accordance with type C requirements of MIL-S-901. The solenoid coil shall be designed for continuous duty and the following:

Voltage rating	As required.
Ambient temperature	50°C. (122°F.).
Temperature rise	Shall not exceed values specified in MIL-C-2212.
Location	Remote.
Enclosure	Spray tight or better.
Terminal block	Required (fitted in the solenoid enclosure).

3.7.7.9 Control circuit transformer. - The transformer shall conform to commercial marine standards and the following:

Volt ampere rating	As required.
Primary voltage, frequency and number of phases	As specified (see 6.1).
Secondary voltage	230 volts, single phase (a.c.).
Enclosure	Spray tight or better.
Voltage regulation	As required.
Duty	Continuous.
Terminal posts	Required.

3.7.7.10 Ignition transformer. - The ignition transformer shall be a unit which has passed Hi-shock tests in accordance with MIL-S-901 and vibration tests in accordance with MIL-STD-167 and which conforms to the following:

Volt ampere rating	As required.
Primary voltage	230 volts, single phase (a.c.).
Secondary voltage	10,000 volts (plus or minus 250 volts).
Enclosure	Spray tight or better for remote mounted units.
Mounting location	Local or remote as required.
Ambient temperature	50°C. (122°F.).
Radio interference suppression	Required.
Duty	Intermittent.

3.7.7.11 Electrical cable, wire and connectors shall conform to MIL-C-915, MIL-C-2194, MIL-C-5015 and MIL-W-16878.

3.7.8 Forced draft blower. - The forced draft blower shall be designed to supply all requisite air to the boiler when steaming at variable loads, through its entire range at a pressure of at least 3.0 inches of water in excess of the maximum pressure required to overcome all resistances of the burner, and the boiler between the air intake of the air regulating control and the exhaust gas outlet of the boiler and with a negative pressure of 3/10 inch water at the suction end. The blower wheel (fan) shall be in satisfactory static and dynamic balance. The interior of the blower casing shall be smooth and free from obstructions such as bolt heads, nuts and pipe fittings. The following construction features shall apply:

(a) The wheel shall be keyed on the shaft.
(b) Provision shall be made for 2-set screws in the hub, located ninety degrees apart.
(c) One set screw shall be located over the key.
(d) One set screw shall be recessed in the shaft for positioning axially.
(e) Frictionless type bearings shall be employed and the shaft shall be provided with a shoulder or collar to prevent axial thrust.
(f) If possible, the blower shall have a direct coupled drive. The maximum usable pressure of the fan shall be determined by the total pressure-volume curve in accordance with the procedures specified in MIL-F-19004.

3.8 Radio interference suppression of the combustion and feedwater control system. - The control circuit, complete, and the circuit component enclosures, wiring conduit, relays, switches, transformers, and similar equipment comprising the complete circuit shall comply with the basic requirements for

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electric power equipments in accordance with MIL-E-917 and as specified for the individual equipment herein. The combustion control system complete with ignition system shall be arranged to minimize radio interference, both radiated and conducted through the use of enclosures, shielded conduits, connectors and line filters. External leads shall be shielded to reduce radio interference by shielded metallic conduit of proper diameter with spherical-seat ball and cone fittings in accordance with MIL-I-16165. All electrical connections and wiring shall be water proof. The measurements and limits specified in MIL-I-16910 shall be used as a guide in designing for radio interference suppression.

3.9 Weight and dimensions.- The physical dimensions and weight of the boiler complete with all auxiliaries, accessories, and controls shall be the minimum commensurate with reliable and efficient operation and shall not exceed the limitations, as specified (see 6.1).

3.10 Identification plates.- Each boiler shall have an identification plate conforming to type A or B of MIL-P-15024, except that identification plates of phenolic compounds are not acceptable. The identification plates shall contain the following information:

(a) Name and description of article.
(b) Serial number.
(c) Federal stock number.
(d) Name and address of manufacturer.
(e) Contract or order number.
(f) A square of about 3/4 inch with letters "INSP" at the top for affixing Government inspector stamp.
(g) The words "U.S."
(h) Design pressure, p.s.i.
(i) Operating pressure, p.s.i.
(j) Hydrostatic test pressure, p.s.i.
(k) Safety valve setting, p.s.i.
(l) Electrical characteristics.

3.10.1 The stock number will be furnished to the contractor by the bureau or agency on written request.

3.11 Repair parts.- Repair parts shall consist of onboard repair parts and stock repair parts.

3.11.1 Onboard repair parts.- The set of onboard repair parts for boiler, and boiler auxiliaries, including electrical equipment, shall be furnished as specified in table II.

Table II- Onboard repair parts

Items	Percent
Boiler parts	
Manhole plate complete with bolts, nuts, and dogs (of each type and size)	100
Manhole gaskets (of each type and size)	800
Handhole plates (of each type and size)	25
Handhole gaskets (of each type and size)	800
Boiler water gage, complete with fittings	50
Boiler water gage glass sets or inserts	200
Safety valve internals (sets) (including seats, discs and springs)	200
Gaskets, complete sets (for boiler and for burner)	800
Drive belt sets (matched) (if used)	200
Burner parts	
Atomizer barrel	200
Sprayer and orifice plates	800
Metering oil valve complete	100
Pilot burner tips (if used)	600

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Table II- Onboard repair parts (cont'd)

Items	Percent
Blower parts	
Bearings set	100
Strainers	
Feed water and fuel oil systems (of each type and size)	200
Fuel oil pump parts	
Pump (complete assembly)	100
Pump relief valve	200
Pump shaft coupling	200
Coupling inserts	600
Seal assemblies	200
Feedwater pump parts	
Shaft (without impeller)	100
Impellers (set)	100
Pump shaft coupling	100
Coupling inserts	600
Packing	200
Gaskets	200
Air pump or compressor (if applicable)	
Air pump assembly	100
Feedwater regulator parts	
Freedwater regulator springs	200
Freedwater regulator packing	200
Control valve complete	100
Low water cutoff control parts	
Float Type	
Switch assembly	100
Float, and rod	100
Magnet (if used)	100
Electrode probe type	
Electrode plugs and probe extensions	400
Resistors	200
Soot blower parts	
Valve discs, valve seats, pinions, and springs as necessary for the type of soot blower furnished	100

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Table II - Onboard repair parts (cont'd)

Items	Percent
Smoke indicator parts (if applicable)	
Optical lens, sealing glass and reflecting mirror as required for the type of smoke indicator	100
Electrical parts	
Motors D.C. one set for each type and size --	
(a) In accordance with MIL-M-17413 for integral H.P. motors including one armature complete with commutator and shaft	100
(b) In accordance with MIL-M-17556 for fractional H.P. motors including one armature complete with commutator and shaft for motors above 1/4 H.P.	100
Motors A.C. one set for each type and size --	
(a) In accordance with MIL-M-17060 for integral H.P. motors	100
(b) In accordance with MIL-M-17059 for fractional H.P. motors including one rotor for motors above 1/4 H.P. (field coils shall not be included)	100
Rotary inverter - unit complete	100
Controller D.C. each type and size used in accordance with MIL-C-2174	100
Controller A.C. each type and size used in accordance with MIL-C-2212	100
Relays of each type and size, including time delay units	100
Ignition transformer	100
Control circuit transformer	100
Ignition electrodes	400
Ignition cables and connectors	400
Combustion safeguard	100
Vacuum tubes of each type and size	400
Combustion safeguard sensing element	400
Modulating motor	100
Oil solenoid valve complete	100
Oil solenoid valve coils	400
Mechanical or electric timers or programmers	100

3.11.1.1 The number of sets of onboard repair parts shall be as specified (see 6.1). Identification marking of repair parts shall be in accordance with MIL-STD-130.

3.11.2 Stock repair parts. - Stock repair parts shall be furnished in accordance with MIL-P-15137.

3.12 Tools. -

3.12.1 A set of special tools and wrenches for required maintenance of the boiler, its auxiliaries and burner assembly shall be provided as specified herein. Tools and wrenches of conventional design are not desired. Each tool shall be indelibly marked to indicate the service for which intended. Special tools are defined as those tools not listed in the Federal Supply Catalog. (This catalog may be consulted in the office of the Government inspector.)

3.12.2 The number of sets of tools and wrenches shall be as specified (see 6.1).

3.12.3 The set of tools and wrenches shall include the following:

1 - Tube expander and mandrel for each size of tube used.
1 - Manhole cover wrench, if required.
1 - Handhole cover wrench, if required.
1 - Ratchet wrench.
1 - Ripping chisel.

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1 - Oyster knife.
1 - Set of tube plugs (blind nipple type, for use with expanders and mandrels furnished for tubes) consisting of 10 percent of the number of each size of tube installed in the boiler.
2 - Safety valve gags.
1 - Safety valve seat and set of disc templates.
1 - Set of gasket templates (one for manhole and one for each size of handhole). The templates shall be of steel 1/16 inch thick.
1 - Set of tube cleaning outfits, air driven in accordance with MIL-C-17370.
1 - Brush set for each 200 generating tubes or fraction thereof (other than A and B rows) of each size installed.
1 - Brush set for each 50 tubes or fraction thereof installed in A and B rows. Brush sets shall be in accordance with MIL-B-2678 and each set shall consist of one complete brush and five refills.
1 - Air lance and 50 foot hose with couplings.
1 - Set of air compressor tools and wrenches as required when compressor is furnished.
1 - Set of feed water pump tools and wrench as required.
1 - Set of soot blower tools and wrenches as required.
1 - Set of tool boxes (metal), as required.
1 - Burner cone form (metal), for use in installing plastic refractory in burner opening.
1 - Burner cone sweep for use with the above form.
1 - Manual ignition torch with torch-holder.

3.13 List of repair parts and tools and wrenches.- Repair parts list shall be in accordance with MIL-P-15137.

3.14 Drawings.- Drawings shall be in accordance with MIL-D-963. The use of manufacturer's standard drawings together with certification data sheets to show design variations is encouraged. Drawings shall contain views, details, data and tabulations as specified for drawings entitled as follows:

(a) Assembly and foundation plan
(1) Assembly views (front, rear and side elevations), plan view and foundation of the boiler complete with all auxiliaries, accessories and controls. All components of the assembly shall be identified by piece number and listed in the material table. Overall dimensions, locations, types and sizes of all terminal connections and space required for tube renewal shall be shown. The following data shall be tabulated:
a. Pressure designed, p.s.i.
b. Pressure operating, p.s.i.
c. Pressure safety valve setting, p.s.i.
d. Pressure hydrostatic test, p.s.i.
e. Heating surface, square feet
f. Furnace volume, cubic feet
g. Evaporating capacity:
1. Rated, pounds per hour
2. Overload, pounds per hour
h. Oil rate:
1. Rated, pounds per hour
2. Overload, pounds per hour
i. Weight dry
j. Weight wet
k. Electrical characteristics
l. Exit gas temperature at overload
m. Efficiency at rated capacity
n. Design rate of combustion KB/HR per square foot of heating surface.
o. Design rate of combustion KB/HR per cubic foot of furnace volume.
(b) Master list of material and tools
(1) List of all items and subassemblies giving number of drawing item breakdown for each subassembly is shown.
(2) List all tools furnished.

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(3) The material table shall be arranged as follows:

Name.
Number required.
Material.
Material specification.
Contractor's drawing number.
Contractor's part number.
Manufacturer's name.
Manufacturer's drawing or part number.
Unit weight.
Remarks.

(c) Boiler construction and weld details-
(1) Section view of the boiler shell assembly as necessary to show arrangement and details of pressure parts, internal fittings and external fittings. Show detail views of all welds. Show method of connecting tubes. Show stress calculations for the pressure vessel.

(d) Boiler sectional views -
(1) Section (side and end) views of the boiler assembly complete with burner. Identify refractory, insulation and lagging; show quantites used and give installation instructions.

(e) Burner assembly and details -
(1) Sectional view of the boiler through the burner. Show and identify all constituent parts. Show detail views of each burner part.

(f) Boiler support foundation -
(1) Plan view, side and end elevation views of the boiler base complete with boiler support members fully dimensioned.

(g) Arrangement and details of boiler casings-

(h) Boiler wiring diagrams-
(1) Show wiring diagram and schematic wiring diagram for the boiler auxiliaries, operating controls and safety controls. Identify all symbols. Identify all components in a material table. Describe, step-by-step, the start-up, operating and securing phases for the boiler control circuit. Describe function of each safety device, how it is integrated into the system and the sequence or change in sequence for each safety shutdown.

(i) Sectional assembly views, as follows – complete with material tables.
(1) Oil pump, complete with performance curves.
(2) Water pumps, complete with performance curves.
(3) Burner-blower, complete with performance curves.
(4) Electric motors, complete with design characteristics and performance curves.
(5) Motor controllers showing design characteristics and the internal wiring circuit.
(6) Belt drive assembly, show arrangement and all design data.
(7) Feedwater regulator and low water cut-off.

3.14.1 All work done prior to approval of drawings shall be entirely at the contractor's risk.

3.14.2 Preliminary drawings.- Preliminary drawings and data shall be submitted. If a vendor proposes to furnish equipment which previously has been supplied and for which the bureau or agency concerned has approved drawings on record, reference to the approved drawings will be acceptable in lieu of preliminary drawings.

3.15 Manuals.- Manuals shall be in accordance with MIL-M-15071. Contents of these manuals shall be arranged in accordance with MIL-M-15071, except that parts identification is not required. In lieu thereof a "List of Components" comprising major parts of the boiler and burner assemblies and all individual auxiliaries, accessories, and controls furnished with the boiler, shall be included. The "List of Components" shall be arranged as follows:

Column 1 - Assembly piece number.
Column 2 - Manufacturer's part number.
Column 3 - Quantity per boiler.
Column 4 - Name in full (abbreviations shall not be used).
Column 5 - Description containing application or function, material, size, rating, manufacturer's type, or model number, catalog number.
Column 6 - Federal stock number (to be added when available).

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Column 7 - Manufacturer's or contractor's drawing number.
Column 8 - Weight.

3.15.1 The list of repair parts and the list of tools and wrenches furnished shall be included in the manual. The arrangement used for these lists shall be the same as specified for the "List of Components".

3.15.2 Reduced size reproductions of the following drawing shall be included in the manual:

(a) Assembly views of the boiler, plan view and foundation.
(b) Sectional views of the boiler.
(c) Arrangement of boiler brickwork.
(d) Arrangement and sectional views of the burner unit.
(e) Assembly and sectional views of all auxiliaries.
(f) Diagrammatic equipment wiring and elementary schematic wiring diagrams.
(g) Master drawings for motors.
(h) Master drawings for controllers.

3.15.3 Manufacturers' standard bulletins applicable to auxiliaries, accessories, and controls furnished shall be included in the manual where necessary for complete description of construction, method of installation, operation, adjustment, and maintenance.

3.15.4 The method of approval of manual shall be as specified in MIL-M-15071, except that two sample copies of the manual complete, shall be submitted for approval to the bureau or agency concerned.

4. QUALITY CONFORMANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction tests. - One boiler shall be submitted to a laboratory satisfactory to the Bureau of Ships for the preproduction tests specified in 4.3.

4.3 Tests. - Tests are required to establish compliance with the requirements of this specification and to ascertain fulfillment of the capacity and efficiency requirements and satisfactoriness of control features and safety devices. Safety devices shall be tested by simulating failure conditions requiring action of each safety device.

4.3.1 Test agenda. - Prior to the start of any testing, a test agenda shall be prepared and submitted to the Bureau of Ships for approval. The agenda shall include all of the tests specified in table III, 4.3.1.1 and the following:

(a) Diagrammatic sketch of the test layout showing location for each reading to be observed.
(b) Instrumentation complete for each item listed in 4.3.1.1 giving make, model and manufacturer of instrument employed.

Table III - Tests

Test	Preproduction unit	Production units
Steady steaming at 120 percent capacity	8-hours consecutively	1-hour
Steady steaming at rated capacity	24-hours consecutively	1-hour
Steady steaming at 50 percent rated capacity	1-hour	1-hour

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Table III - Tests (cont'd)

Test	Preproduction unit	Production units
Steady steaming at minimum capacity	8-hours	1-hour
Steaming at loads modulated from 120 percent capacity to minimum capacity	8-hours consecutively	---
Steaming at low loads "On-Off" cycling	300-cycles consecutively	10-cycles consecutively
Safety valve by steam	1-test	1-test
Combustion safeguard (Steaming boiler)	10-cycles consecutively	5-cycles consecutively
Low water cut-off (Steaming boiler)	10-cycles consecutively	5-cycles consecutively
Air-pressure cut-off	10-cycles consecutively	5-cycles consecutively
Low voltage	1-cycle	1 cycle
Hydrostatic test 150 percent design p.s.i.	1 Test	1 Test
Boiler weight wet/dry	Once	Once
Chemical and physical properties of oil	1-sample	---

4.3.1.1 During the operational runs specified in table III the following data shall be obtained and recorded every 15 minutes, except for the steady steaming run on the preproduction unit where data shall be obtained and recorded once each hour. Efficiency of the boiler shall be calculated. Performance curves of the boiler shall be plotted.

(a) Fuel oil consumption
(b) Fuel oil supply pressure
(c) Return line oil pressure
(d) Steam pressure
(e) Steam quality
(f) Blower air inlet temperature[1/]
(g) Blower inlet pressure or barometric pressure (as applicable)[1/]
(h) Blower discharge pressure[1/]
(i) Burner plenum pressure
(j) Furnace pressure (measured at front center and rear of furnace on centerline, and on both side walls)
(k) Gas temperature at breeching outlet[1/ 2/]
(l) Gas pressure at breeching outlet[1/]
(m) Boiler casing temperatures[1/]
(n) Gas analyses CO_2, O_2, and CO at boiler outlet (percent)[2/]
(o) Water input or steam output rate
(p) Feedwater pump discharge pressure
(q) Feedwater temperature
(r) Water level in gage glass
(s) Motor amperes
(t) Line voltage
(u) Total electric power consumption
(v) RPM of each drive and driven.

1/ Not required on the production units.

2/ Average readings - Sufficient readings shall be taken at various depths across a plane normal to the gas flow to give representative average values. Test agenda shall include procedure for obtaining these readings and shall show by sketch the intended sampling points and probe locations.

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4.4 Production inspection.- Each boiler offered for delivery shall be tested as specified in 4.3 and as follows:

(a) Examined to determine compliance with the approved drawings.
(b) Examined to determine compliance with section 5 herein.
(c) Where other specifications form a part of this specification, sampling, examination and tests shall be conducted as required in the referenced specification.

4.4.1 Certified report.- Four copies of a certified report covering the production inspection specified in 4.3 and 4.4 shall be furnished the bureau or agency concerned.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

5.1.1 Basic equipment.-

5.1.1.1 Preservation and packaging.- Preservation and packaging of the boiler, detached auxiliaries and accessories, shall be sufficient to afford adequate protection against corrosion, deterioration, physical and mechanical damage during shipment from the supply source to the using activity until early installation and may conform to the supplier's commercial practice when such meets these requirements. All openings shall be sealed to prevent entrance of foreign matter.

5.1.1.2 Packing.- The boiler shall be shipped assembled as a complete unit, except such auxiliaries and accessories that require separate packing for preservation, packaging and security in shipment. Such parts shall be installed by the shipyard. Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, contract or other number, contractor's name and destination.

5.1.2 Onboard repair parts and tools.- Onboard repair parts and tools shall be preserved and packaged level A, packed level C and marked levels A and C in accordance with the applicable specifications as follows:

(a) Mechanical parts and tools.- MIL-R-196 and MIL-P-17286.
(b) Electrical parts.- MIL-P-16298 and MIL-E-17555.

5.1.2.1 Items not specifically covered by the specifications referenced in 5.1.2 shall be unit protected and packaged in accordance with MIL-P-116. Repair parts and tools shall be individually packaged, except when used in sets.

5.2 Domestic shipment and storage or overseas shipment requirements.- The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging.-

5.2.1.1.1 The boiler and accessories shall be preserved and packaged in accordance with level A requirements of MIL-B-3180.

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5.2.1.2 Repair parts and tools.- Repair parts and tools shall be preserved and packaged level A in accordance with MIL-R-196, MIL-P-17286, MIL-P-16298 and MIL-E-17555 as applicable. Items not covered therein shall be unit protected and packaged in accordance with the guide lines of MIL-P-116. Unless otherwise specified parts shall be individually packaged except when used in sets.

5.2.1.3 Packing.-

5.2.1.3.1 Equipment.-

5.2.1.3.1.1 Level A.- Boilers shall be packed in sheathed crates conforming to MIL-C-104.

5.2.1.3.1.1.1 Detached boiler auxiliaries, accessories and controls which are shipped (knocked down) for installation shall be packed in overseas type wood cleated plywood, nailed wood or fiber boxes conforming to PPP-B-601, PPP-B-621 or PPP-B-636 (class 2), respectively, at the option of the contractor. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds. Single items having a gross weight exceeding 200 pounds shall be packed in boxes conforming to PPP-B-601 or PPP-B-621, modified in accordance with the applicable box specification. Boxes shall be lined with a sealed caseliner conforming to MIL-L-10547.

5.2.1.3.1.2 Level B.- Boilers shall be packed in sheathed crates conforming to MIL-C-104. Boilers having a gross weight less than 10,000 pounds may be individually packed in unsheathed crates conforming to MIL-C-3774 with the following additional requirements:

(a) The burner or front end of the boiler shall be completely enclosed in a lumber or plywood housing. The housing shall be anchored to the skid base of the crate and securely strapped to the boiler.

(b) The entire contents of the crate shall be shrouded approximately two-thirds of the way down with waterproof barrier material conforming to UU-P-271. The shrouding shall be securely fastened and arranged in a manner to prevent the formation of water pockets.

5.2.1.3.1.2.1 Detached boiler auxiliaries, accessories and controls which are shipped (knocked down) for installation shall be packed as specified in 5.2.1.3.1.1.1. Boxes shall be of the domestic type or class. Caseliners shall not be required.

5.2.1.3.2 Repair parts and tools.- Repair parts and tools shall be packed level A or B as specified in accordance with the applicable specification specified in 5.2.1.2.

5.2.1.3.3 Anchoring, blocking, bracing and cushioning shall be in accordance with JAN-P-100 and the appendix to MIL-C-104.

5.2.1.4 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.)

5.2.1.4.1 Unpacking instructions shall be provided and shall be sealed in waterproof envelopes which shall be tacked, stapled or otherwise securely attached to the outside of each basic equipment container in the most protected location. The words "UNPACKING INSTRUCTIONS" shall be stenciled on each container indicating location of such instructions.

5.2.1.5 Manuals.- Manuals shall be packaged and packed for the level of protection specified in accordance with MIL-M-15071).

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Specify and define hand of the boiler (see 3.3.2).

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(c) Capacity of boiler in pounds per hour (see 3.3.4).

(1) Normal _____
(2) Maximum _____

(d) Fuel oil to be burned (see 3.3.4 and 3.6.2).
(e) Type of burner operation (manual or automatic) (see 3.3.5).
(f) Electric power supply available (see 3.3.6.3 and 3.7.7.1.1).
(g) Primary voltage, frequency and number of phases for control circuit transformer (see 3.7.7.9).
(h) Limiting physical dimensions and weight of boiler assembly complete with all auxiliaries, accessories, and controls (see 3.9).
(i)

(1) Overall height to the top of safety valve pad.
(2) Overall width.
(3) Depth over complete assembly.
(4) Floor space occupied by the boiler assembly complete with all auxiliaries.
(5) Dry weight of the boiler assembly complete with all auxiliaries, accessories, and controls.

(j) Number of sets of onboard repair parts required (see 3.11.1.1).
(k) Selection of packaging, packing, or marking requirements if other than as specified in 5.1 (see 5.2).

6.2 Bid data.- The bidder shall furnish, with their proposals, the following data in triplicate:

(a) Rated capacity of the boilers, pounds steam/hour, normal and maximum at design conditions (see 3.3.4).
(b) Heating surface square feet.
(c) Furnace volume cubic feet.
(d) Anticipated performance data at the rated normal and maximum capacities (efficiency, CO_2 content and temperature of exit gas and pounds of oil per hour based on burning diesel fuel conforming to MIL F-16884 and JP-5 fuel conforming to MIL-J-5624).
(e) Complete description of the boiler, burner and auxiliaries.
(f) Complete description of controls and safety devices (manufacturer's bulletins shall be attached).
(g) Weight, dry and wet of the complete unit with attached auxiliaries.
(h) Itemized list of proposed onboard repair parts.
(i) Itemized list of tools and wrenches, special.
(j) Estimated weights of onboard repair parts and tools and wrenches.
(k) Two sets of blueprints of the following preliminary drawings:

(1) The outline assembly views (front, sides and rear) of the boiler complete with auxiliaries, accessories, and controls giving overall dimensions, locations of all piping connections, safety devices, controls, and space required for tube renewal.
(2) Sectional elevation and plan views of the boiler and burner assembly showing arrangement of tubes and internal fittings and giving sizes of boiler tubes, thickness of shell plate, and identifying refractories and insulation used.

(l) Satisfactory evidence to show that the bidder has had adequate experience in manufacture of boilers of the type and size offered in the bid.

6.2.1 Right is reserved to reject bids lacking in full or part the data required in 6.2.

6.3 Preproduction.- Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to

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rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Preparing activity:
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Project 4410-N023 Sh

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

BUDGET BU. NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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